LUCIDMSK CME PROGRAM

MODULE 02 — POST-TEST

Knee MRI: Grading, Measurements & Instability

Knee · 10 Questions · 1.0 AMA PRA Cat 1 Credit(s)™

Read each question carefully and select the single best answer. A score of ≥ **70%** (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1 **Partial cartilage loss >50% depth over a 2 cm area of the medial femoral condyle WITHOUT subchondral bone exposure. Modified Outerbridge:**

- **A** Grade I
- **B** Grade II
- **C** Grade III
- **D** Grade IV

2 **A vertical peripheral meniscal tear in the red-red zone has the highest potential for:**

- **A** Partial meniscectomy
- **B** Surgical repair
- **C** Observation only
- **D** Transplant

3 **Which sign is PATHOGNOMONIC for ACL rupture?**

- **A** Posterior tibial bone contusion
- **B** Segond fracture
- **C** PCL buckling
- **D** Joint effusion

4 **TT-TG distance is 22 mm. Dejour Type B dysplasia is present. Most appropriate surgical consideration:**

- **A** Lateral release alone
- **B** MPFL reconstruction alone
- **C** Tibial tubercle medialization + MPFL
- **D** Conservative management

5 Greif Grade 3 ramp lesion on MRI:

A	No signal abnormality
B	Partial tear; no gap
C	Complete non-displaced tear
D	Displaced/fluid-distended gap at posteromedial meniscocapsular junction

6 Insall-Salvati ratio is 1.5. This indicates:

A	Patella baja
B	Normal patellar height
C	Patella alta
D	Trochlear dysplasia only

7 Dejour Type D trochlear dysplasia has which combination of lateral X-ray signs?

A	Crossing sign only
B	Crossing sign + supratrochlear spur
C	Crossing sign + double contour
D	All three: crossing sign + spur + double contour

8 Complete MCL tear (Grade III) on coronal MRI shows:

A	Periligamentous edema; continuous fibers
B	Partial discontinuity; T2 within ligament
C	Complete fiber discontinuity; fluid gap; wavy ends
D	Thickening without signal change

9 Normal posterior tibial slope on sagittal MRI:

A	0–3°

B	7–10°
C	15–18°
D	>20°

10 **A horizontal cleavage tear splits the meniscus into superior and inferior leaves. Most likely management:**

A	Observation
B	Primary suture repair
C	Partial meniscectomy (often superior leaf)
D	Total meniscectomy

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	C	B	B	C	D	C	D	C	B	C

Q1 **Partial cartilage loss >50% depth over a 2 cm area of the medial femoral condyle WITHOUT subchondral bone exposure. Modified Outerbridge:** **Answer: C**

A Grade I

B Grade II

✓ **Grade III**

D Grade IV

EXPLANATION Grade III = >50% depth partial loss, defect >1.5 cm, no bone exposure. Grade II = partial <50%. Grade IV = bone exposed. Grade I = signal change with intact surface.

Q2 **A vertical peripheral meniscal tear in the red-red zone has the highest potential for:** **Answer: B**

A Partial meniscectomy

✓ **Surgical repair**

C Observation only

D Transplant

EXPLANATION Red-red zone vertical longitudinal tears have excellent vascularity and are the best candidates for meniscal suture repair rather than resection.

Q3 **Which sign is PATHOGNOMONIC for ACL rupture?** **Answer: B**

A Posterior tibial bone contusion

✓ **Segond fracture**

C PCL buckling

D Joint effusion

EXPLANATION The Segond fracture (lateral tibial rim avulsion) is pathognomonic for ACL tear. The others are secondary supportive signs but are not pathognomonic.

Q4 **TT-TG distance is 22 mm. Dejour Type B dysplasia is present. Most appropriate surgical consideration:** **Answer: C**

A Lateral release alone

B MPFL reconstruction alone

✓ **Tibial tubercle medialization + MPFL**

D Conservative management

EXPLANATION TT-TG >20 mm = threshold for tibial tubercle medialization. Major dysplasia (Dejour B/C/D) may require trochleoplasty. MPFL reconstruction is combined with bony procedures.

Q5 **Greif Grade 3 ramp lesion on MRI:** **Answer: D**

A No signal abnormality

B Partial tear; no gap

C Complete non-displaced tear

✓ **Displaced/fluid-distended gap at posteromedial meniscocapsular junction**

EXPLANATION Greif Grade 3 = displaced or fluid-distended gap = frank unstable ramp lesion requiring repair. Grade 2 = complete but non-displaced. Grade 1 = partial without full discontinuity.

Q6 **Insall-Salvati ratio is 1.5. This indicates:** **Answer: C**

A Patella baja

B Normal patellar height

✓ **Patella alta**

D Trochlear dysplasia only

EXPLANATION Insall-Salvati >1.2 = patella alta. Normal = 0.8–1.2. Patella baja = <0.8. Alta reduces trochlear engagement early in flexion, predisposing to patellar instability.

Q7 **Dejour Type D trochlear dysplasia has which combination of lateral X-ray signs?** **Answer: D**

A Crossing sign only

B Crossing sign + supratrochlear spur

C Crossing sign + double contour

✓ **All three: crossing sign + spur + double contour**

EXPLANATION Type D (most severe) = all three signs: crossing sign + supratrochlear spur + double contour (cliff pattern). Asymmetric trochlea on axial MRI. Types B–D = major dysplasia.

Q8 **Complete MCL tear (Grade III) on coronal MRI shows:** **Answer: C**

A Periligamentous edema; continuous fibers

B Partial discontinuity; T2 within ligament

✓ **Complete fiber discontinuity; fluid gap; wavy ends**

D Thickening without signal change

EXPLANATION Grade III = complete fiber discontinuity with fluid gap, wavy/retracted ends. Grade I = edema, intact fibers. Grade II = partial discontinuity. Always assess for concurrent posterolateral corner injury.

Q9 **Normal posterior tibial slope on sagittal MRI:** **Answer: B**

A 0–3°

✓ **7–10°**

C 15–18°

D >20°

EXPLANATION Normal posterior slope 7–10°. Increased slope increases ACL loading — relevant in ACL revision surgery and slope-reducing osteotomy (tibial deflexion osteotomy).

Q10 **A horizontal cleavage tear splits the meniscus into superior and inferior leaves. Most likely management:** **Answer: C**

A Observation

B Primary suture repair

✓ **Partial meniscectomy (often superior leaf)**

D Total meniscectomy

EXPLANATION Horizontal cleavage tears generally have poor repair potential due to limited vascularity. Partial meniscectomy of the unstable leaf is most common; repair is possible in young patients with peripheral tears.